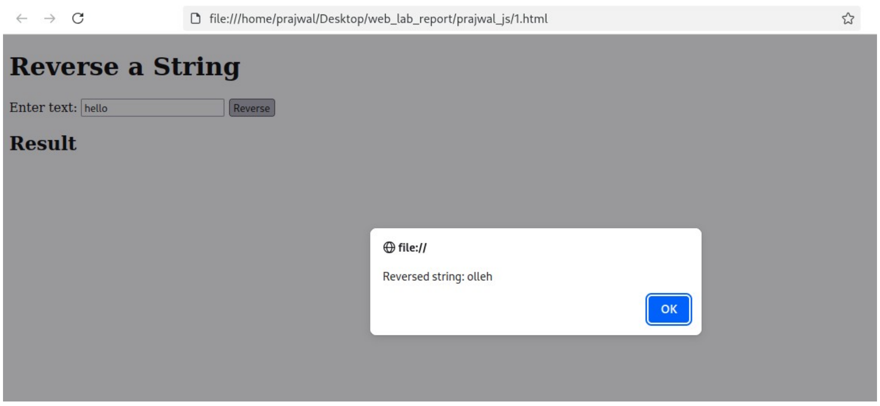
Lab works: JavaScript

1. **Write a JavaScript program to take a string input from the user and display it in reverse order.**

```
<!DOCTYPE html>
<html lang="en">
<head>
	<meta charset="utf-8" />
	<meta name="viewport" content="width=device-width, initial-scale=1" />
	<title>Reverse String</title>
	<script>
		function reverseString(str) {
			return String(str).split('').reverse().join('');
		}

		function onReverseClick() {
			const input = document.getElementById('textInput').value;
			const reversed = reverseString(input);
			document.getElementById('result').textContent = reversed;
			alert('Reversed string: ' + reversed);
		}
	</script>
</head>
<body>
	<h1>Reverse a String</h1>
	<label for="textInput">Enter text:</label>
	<input id="textInput" type="text" placeholder="Type something" />
	<button onclick="onReverseClick()">Reverse</button>

	<h2>Result</h2>
	<div id="result" style="font-weight:600;color:#2a2a2a"></div>
</body>
</html>
```



2. Write a JavaScript program that accepts a number and

- **Finds its reverse**
- **Checks if it is a palindrome**
- **Checks if it is an Armstrong number**

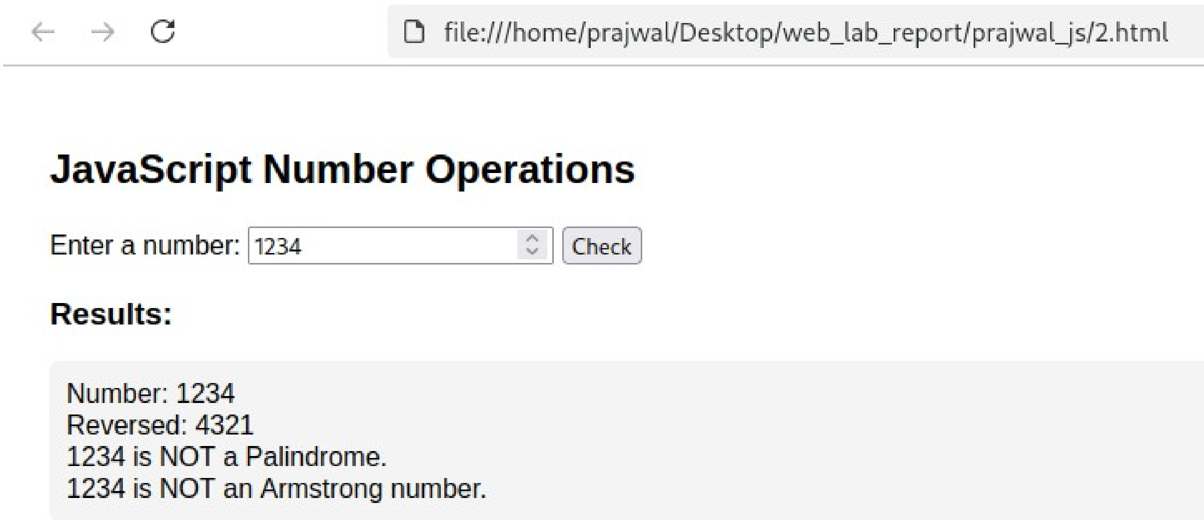
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Number Check: Reverse, Palindrome, Armstrong</title>
</head>
<body style="font-family: Arial, sans-serif; padding: 20px;">
  <h2>JavaScript Number Operations</h2>
  <label for="num">Enter a number:</label>
  <input type="number" id="num" />
  <button onclick="checkNumber()">Check</button>
  <h3>Results:</h3>
  <div id="output" style="white-space: pre-line; font-size: 16px; background: #f4f4f4; padding: 10px; border-radius: 5px;"></div>
<script>
    function reverseNumber(num) {
      let reversed = 0;
      let original = num;
     while (num > 0) {
        let digit = num % 10;
        reversed = reversed * 10 + digit;
        num = Math.floor(num / 10);
      }
      return reversed;
    }
```

```
function isPalindrome(num) {
  return num === reverseNumber(num);
}
function isArmstrong(num) {
  let sum = 0;
  let temp = num;
  let digits = num.toString().length;
  while (temp > 0) {
    let digit = temp % 10;
    sum += Math.pow(digit, digits);
    temp = Math.floor(temp / 10);
  }
  return sum === num;
}
function checkNumber() {
  let num = parseInt(document.getElementById("num").value);
  let output = "";

  if (isNaN(num)) {
    document.getElementById("output").innerText = "⚠ Please enter a
valid number.";
    return;
  }      output += "Number: " + num + "\n";
  output += "Reversed: " + reverseNumber(num) + "\n"
  if (isPalindrome(num)) {
    output += num + " is a Palindrome.\n";
  } else {
    output += num + " is NOT a Palindrome.\n";
  }
  if (isArmstrong(num)) {
    output += num + " is an Armstrong number.\n";
  } else {
```

```
        output += num + " is NOT an Armstrong number.\n";
      }

      document.getElementById("output").innerText = output;
    }
  </script>
</body>
</html>
```

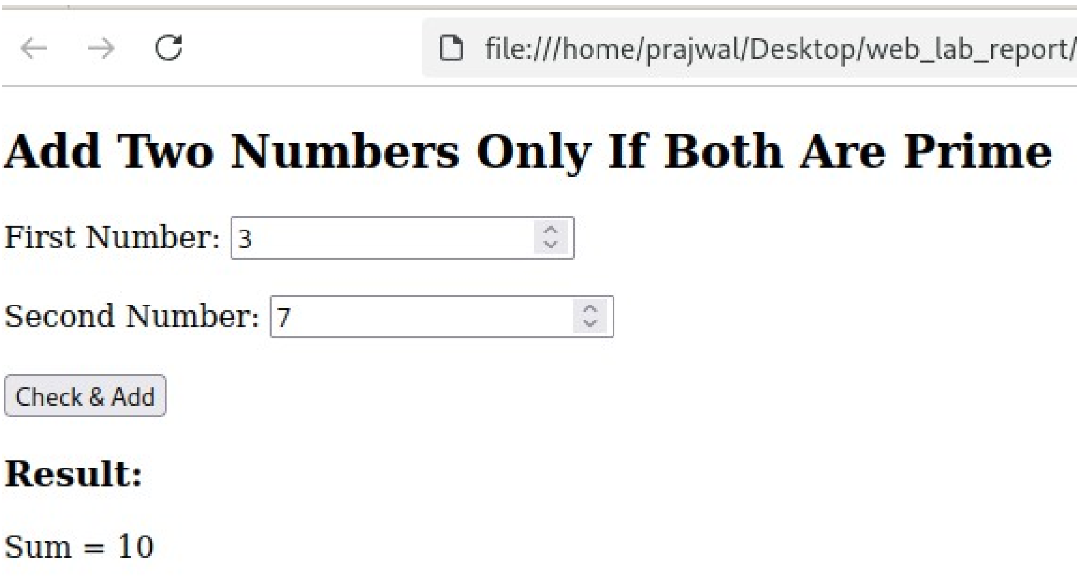


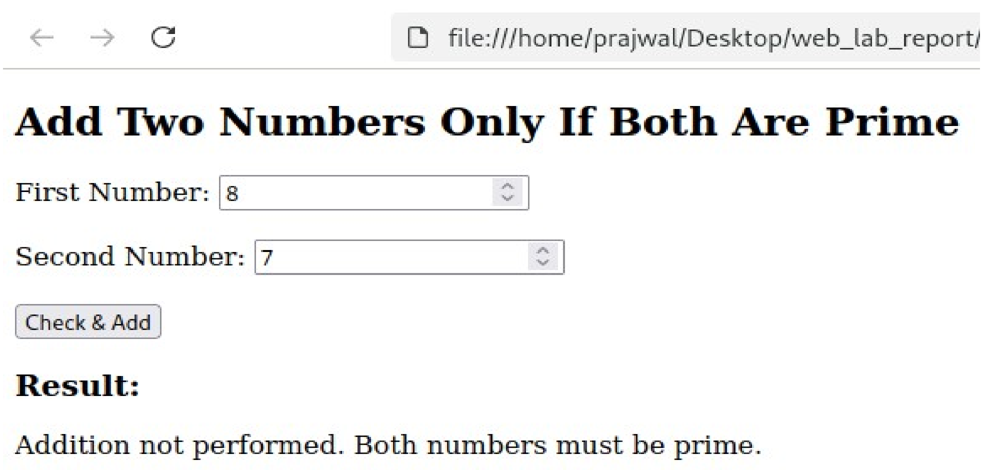
3. Write a JavaScript program to add two numbers only if both are prime. Use a user defined function to check for primality before performing the addition.

```
 <!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Add Two Prime Numbers</title>
</head>
<body>
  <h2>Add Two Numbers Only If Both Are Prime</h2>
  <label>First Number: </label>
  <input type="number" id="num1"><br><br>
  <label>Second Number: </label>
  <input type="number" id="num2"><br><br>
  <button onclick="addIfPrime()">Check & Add</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    function isPrime(n) {
      if (n < 2) return false;
      for (let i = 2; i <= Math.sqrt(n); i++) {
        if (n % i === 0) return false;
      }
      return true;
    }
    function addIfPrime() {
      let a = parseInt(document.getElementById("num1").value);
      let b = parseInt(document.getElementById("num2").value);

      if (isPrime(a) && isPrime(b)) {
        document.getElementById("output").innerText = "Sum = " + (a + b);
```

```
    } else {
      document.getElementById("output").innerText = "Addition not performed. Both numbers must be prime.";
    }
  }
</script>
</body>
</html>
```





4. Write a JavaScript program that:

- **Adds two numbers if both are even**
- **Otherwise, finds their difference (Use user-defined functions for the checks.)**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Add if Even, Otherwise Difference</title>
</head>
<body>
  <h2>JavaScript: Add if Even, Otherwise Find Difference</h2>
  <label>First Number: </label>
  <input type="number" id="num1"><br><br>
  <label>Second Number: </label>
  <input type="number" id="num2"><br><br>
  <button onclick="calculate()">Calculate</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    function isEven(n) {
      return n % 2 === 0;
    }
    function calculate() {
      let a = parseInt(document.getElementById("num1").value);
      let b = parseInt(document.getElementById("num2").value);

      if (isEven(a) && isEven(b)) {
        document.getElementById("output").innerText = "Sum = " + (a + b);
      } else {
        document.getElementById("output").innerText = "Difference = " +
Math.abs(a - b);
```

```
      }
    }
  </script>
</body>
</html>
```

5. Write a JavaScript program to compare two strings.

- **If they are equal, display: "The first string is in uppercase"**
- **Otherwise, display: "The strings are not equal"**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Compare Two Strings</title>
</head>
<body>
  <h2>JavaScript: Compare Two Strings</h2>
  <label>First String: </label>
  <input type="text" id="str1"><br>
  <label>Second String: </label>
  <input type="text" id="str2"><br>
  <button onclick="compareStrings()">Compare</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    function compareStrings() {
      let s1 = document.getElementById("str1").value;
      let s2 = document.getElementById("str2").value;
      if (s1 === s2) {
        document.getElementById("output").innerText = "The first string is in uppercase";
      } else {
        document.getElementById("output").innerText = "The strings are not equal";
      }
    }
  </script>
</body>
```

</html>

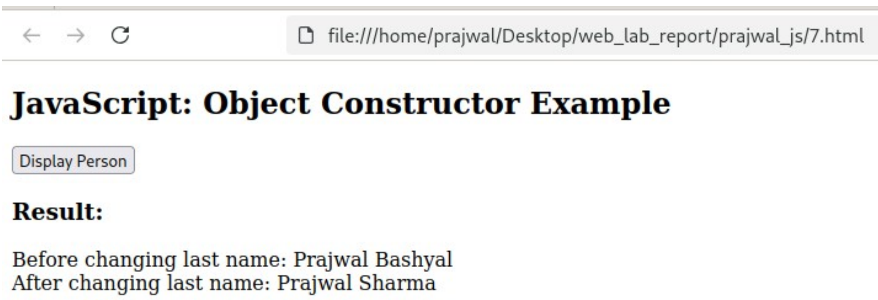

file:///home/prajwal/Desktop/web_lab_report/prajwal_js/5.html
JavaScript: Compare Two Strings
First String: prajwal
Second String: prajwal
Compare
Result:
The first string is in uppercase

6. Create an associative array in JavaScript and display its content using a for...in loop.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Associative Array in JavaScript</title>
</head>
<body>
  <h2>JavaScript: Associative Array Example</h2>
  <button onclick="showArray()">Display Array</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
   function showArray() {
     let person = {name: "John", age: 25, city: "New York"};
     let result = "";
     for (let key in person) {
       result += key + " : " + person[key] + "\n";
     }
     document.getElementById("output").innerText = result;
   }
  </script>
</body></html>
```

7. Write a JavaScript program using an object constructor to set a person's first name and last name. Add a method to change the last name. Display the name before and after calling the method.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Object Constructor Example</title></head>
<body>
  <h2>JavaScript: Object Constructor Example</h2>
  <button onclick="showPerson()">Display Person</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    function Person(firstName, lastName) {
      this.firstName = firstName;
      this.lastName = lastName;
      this.changeLastName = function(newLastName) {
        this.lastName = newLastName;
     }  }
    function showPerson() {
      let p = new Person("John", "Doe");
      let output = "Before changing last name: " + p.firstName + " " + p.lastName + "\n";
      p.changeLastName("Smith");
      output += "After changing last name: " + p.firstName + " " + p.lastName;
      document.getElementById("output").innerText = output;}
     </script>  </body>
    </html>
```



8. Write a JavaScript program to display a real-time clock in the format Hour:Minute:Second.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Real-Time Clock</title>
</head>
<body>
  <h2>JavaScript: Real-Time Clock</h2>
  <div id="clock" style="font-size: 24px; font-weight: bold;"></div>
  <script>
    function updateClock() {
      let now = new Date();
      let h = now.getHours().toString().padStart(2, '0');
      let m = now.getMinutes().toString().padStart(2, '0');
      let s = now.getSeconds().toString().padStart(2, '0');
      document.getElementById("clock").innerText = h + ":" + m + ":" + s;
    }
    setInterval(updateClock, 1000);
    updateClock();
  </script>
</body>
</html>
```

9. Use the Date object to compare today’s date with February 15,2015. Compare and Alert whether today’s date is beforeor after the given date.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Compare Dates</title>
</head>
<body>
  <h2>JavaScript: Compare Today's Date with Feb 15, 2015</h2>
  <button onclick="compareDate()">Compare Dates</button>
  <script>
    function compareDate() {
      let today = new Date();
      let givenDate = new Date(2015, 1, 15); // Month is 0-indexed: 0=Jan, 1=Feb
      if (today < givenDate) {
        alert("Today's date is before February 15, 2015.");
      } else if (today > givenDate) {
        alert("Today's date is after February 15, 2015.");
      } else {
        alert("Today's date is February 15, 2015.");
      }   }
  </script>
</body>
</html>
```

10.Define a Car class in JavaScript.

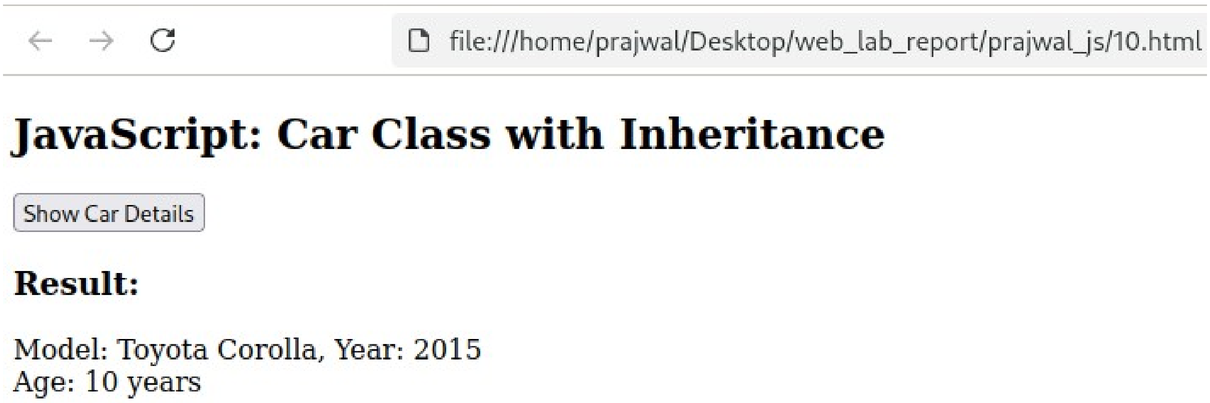
▪ Set properties such as model and manufacturing year.

▪ Display these properties.

▪ Use inheritance to calculate the car's age based on the current year

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Car Class Example</title>
</head>
<body>
  <h2>JavaScript: Car Class with Inheritance</h2>
  <button onclick="showCar()">Show Car Details</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    class Car {
      constructor(model, year) {
        this.model = model;
        this.year = year;}
      display() {
        return "Model: " + this.model + ", Year: " + this.year;
      }}
  class CarAge extends Car {
      calculateAge() {
        let currentYear = new Date().getFullYear();
        return currentYear - this.year;
      } }
    function showCar() {
      let myCar = new CarAge("Toyota Corolla", 2015);
      let output = myCar.display() + "\nAge: " + myCar.calculateAge() + " years";
```

```
      document.getElementById("output").innerText = output; }
  </script>
</body>
</html>
```



11. Write JavaScript programs to handle errors using:

a) try...catch statement

b) window.onerror event

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>JavaScript Error Handling</title>
</head>
<body>
  <h2>JavaScript: Error Handling Examples</h2>
  <button onclick="testTryCatch()">Test try...catch</button>
  <button onclick="testError()">Test window.onerror</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    // a) Using try...catch
    function testTryCatch() {
      try {
        let x = y + 5; // y is undefined, will throw error
      } catch (err) {
        document.getElementById("output").innerText = "Caught an error: " + err;
      }  }
   // b) Using window.onerror
    window.onerror = function(message, source, lineno, colno, error) {
      document.getElementById("output").innerText = "Global error caught: " +
message;
      return true; // prevent default browser error alert
    } function testError() {
      z + 10; // z is undefined, triggers window.onerror }
  </script>
</body>
```

</html>


File
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Error Handling in JavaScript
a) try...catch
Click the button to run code that throws an error; it will be caught by try...catch.
Run try...catch demo
b) window.onerror
Click the button to trigger an uncaught error that is handled by window.onerror.
Trigger global error

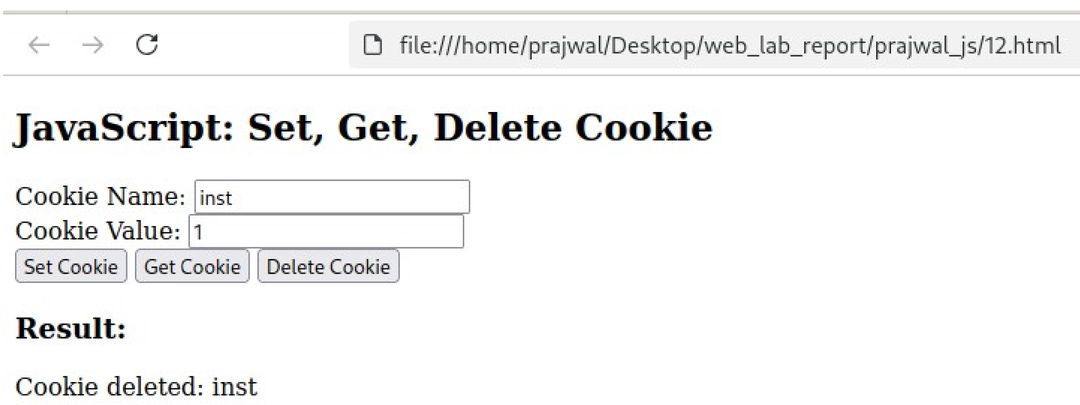
12. Write a JavaScript program to set a cookie, get a cookie, and delete a cookie.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Cookie Operations in JavaScript</title>
</head>
<body>
  <h2>JavaScript: Set, Get, Delete Cookie</h2>
  <label>Cookie Name: </label><input type="text" id="cname"><br>
  <label>Cookie Value: </label><input type="text" id="cvalue"><br>
  <button onclick="setCookie()">Set Cookie</button>
  <button onclick="getCookie()">Get Cookie</button>
  <button onclick="deleteCookie()">Delete Cookie</button>
  <h3>Result:</h3>
  <div id="output"></div>
  <script>
    function setCookie() {
      let name = document.getElementById("cname").value;
      let value = document.getElementById("cvalue").value;
      document.cookie = name + "=" + value + "; path=/";
      document.getElementById("output").innerText = "Cookie set: " + name +
"=" + value;
    }

    function getCookie() {
      let name = document.getElementById("cname").value + "=";
      let decodedCookie = decodeURIComponent(document.cookie);
      let ca = decodedCookie.split(';');
      for(let i = 0; i < ca.length; i++) {
        let c = ca[i].trim();
        if (c.indexOf(name) === 0) {
```

```
        document.getElementById("output").innerText = "Cookie value: " +
c.substring(name.length);
        return;
      }
    }
    document.getElementById("output").innerText = "Cookie not found";
  }

  function deleteCookie() {
    let name = document.getElementById("cname").value;
    document.cookie = name + "=; expires=Thu, 01 Jan 1970 00:00:00 UTC;
path=/;";
    document.getElementById("output").innerText = "Cookie deleted: " + name;
  }
 </script>
</body>
</html>
```



13. Create a Form and write a JavaScript program to validate the following user inputs:

■ Email–must be valid email and ends with @gmail.com or @hotmail.com

■ Phone number–accept if it is either:

■ NTC: 9841****** or 9851******

■ NCELL: 980******* ■

Password–accept if it contains:

■ At least one capital letter

■ At least one small letter

■ At least one number

■ At least one special symbol

■ Minimum length of 12 characters

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Form Validation</title>
</head>
<body>
  <h2>JavaScript Form Validation</h2>
  <form id="myForm" onsubmit="return validateForm()">
    <label>Email: </label>
    <input type="text" id="email"><br>
    <label>Phone Number: </label>
    <input type="text" id="phone"><br>
    <label>Password: </label>
    <input type="password" id="password"><br>
    <button type="submit">Submit</button>
  </form>
  <h3>Result:</h3>
  <div id="output"></div><script>
```

```
    function validateForm() {
      let email = document.getElementById("email").value;
      let phone = document.getElementById("phone").value;
      let password = document.getElementById("password").value;
      let output = "";
      // Email validation
      let emailPattern = /^[a-zA-Z0-9._%+-]+@(gmail\.com|hotmail\.com)$/;
      if (!emailPattern.test(email)) {
        output += "Invalid email. Must end with @gmail.com or @hotmail.com.\n";
      }  // Phone validation
      let phonePattern = /^(9841\d{6}|9851\d{6}|980\d{7})$/;
      if (!phonePattern.test(phone)) {
        output += "Invalid phone number. Must be NTC: 9841****** or 9851******,
NCELL: 980*******\n";
      } // Password validation
      let passwordPattern = /^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[\W_]).{12,}$/;
      if (!passwordPattern.test(password)) {
        output += "Invalid password. Must have at least one uppercase, one
lowercase, one number, one special symbol, and minimum 12 characters.\n";
      }
 if (output === "") {
        output = "All inputs are valid!";      }
   document.getElementById("output").innerText = output;
      return false; // prevent actual form submission   }
  </script>
</body>
</html>
```

jQuery Lab Questions

14.Write a jQuery program to toggle the background color of a webpage. Also, toggle a paragraph’s visibility when a button is clicked.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>jQuery Toggle Example</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
</head>
<body>
  <h2>jQuery: Toggle Background and Paragraph</h2>
  <button id="toggleBtn">Toggle</button>
  <p id="myPara">This is a paragraph to show/hide.</p>
  <script>
    $(document).ready(function() {
      $("#toggleBtn").click(function() {
        // Toggle background color
        if ($("body").css("background-color") === "rgb(255, 255, 255)") {
          $("body").css("background-color", "lightblue");
        } else {
          $("body").css("background-color", "white");   }
        // Toggle paragraph visibility
        $("#myPara").toggle();
      });
    });
  </script>
</body>
</html>
```


file:///home/prajwal/Desktop/web_lab_report/prajwal_js/14.html
jQuery: Toggle Background and Paragraph
Toggle


file:///home/prajwal/Desktop/web_lab_report/prajwal_js/14.html
jQuery: Toggle Background and Paragraph
Toggle
This is a paragraph to show/hide.

15.Write a jQuery program to animate a car with options to move forward, stop, and reverse.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Car Animation</title>
  <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
  <style>
    #car { width: 100px; height: 50px; background: red; position: absolute; top: 100px; left: 0; }
  </style>
</head>
<body>
  <h2>jQuery Car Animation</h2>
  <button id="forward">Move Forward</button>
  <button id="stop">Stop</button>
  <button id="reverse">Reverse</button>
  <div id="car"></div>
  <script>
    let anim;
    $("#forward").click(function() {
      $("#car").stop();
      anim = $("#car").animate({left: "+=500px"}, 5000);
    });
    $("#reverse").click(function() {
      $("#car").stop();
      anim = $("#car").animate({left: "-=500px"}, 5000);
    });
    $("#stop").click(function() {
      $("#car").stop();
    });
  </script>
```

```
</body>
</html>
```


file:///home/prajwal/Desktop/web_lab_report/prajwal_js/15.html
jQuery Car Animation
Move Forward
Stop
Reverse

AJAX Lab Questions

16.Write an AJAX program to fetch a JSON data object, parse it, and display its contents in HTML using a card layout.

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>AJAX JSON Example - Nepali Names</title>

  <style>

    .card { border: 1px solid #ccc; padding: 10px; margin: 10px; border-radius: 5px; width: 200px; display: inline-block; }

  </style>

</head>

<body>

  <h2>AJAX: Display Nepali Names as Cards</h2>

  <button onclick="fetchData()">Show Names</button>

  <div id="output"></div>

  <script>

    function fetchData() {

      // Sample Nepali JSON data

      let data = [

        {name: "Prajwal", email: "prajwal@example.com", city: "Butwal"},

        {name: "Roshan", email: "roshan@example.com", city: "Kathmandu"},

        {name: "Prince", email: "prince@example.com", city: "Pokhara"},

        {name: "Ayush", email: "ayush@example.com", city: "Lalitpur"},

        {name: "Sita", email: "sita@example.com", city: "Biratnagar"}

      ];

      let output = "";

      data.forEach(user => {

        output += `<div class="card"><strong>${user.name}</strong><br>Email: ${user.email}<br>City: ${user.city}</div>`;

      });
```

```
        document.getElementById("output").innerHTML = output;
      }
    </script>
  </body>
  </html>
```

17.Write an AJAX program to read a file asynchronously and render its contents in HTML.

•Use proper server response handling.

•Process the file only if the status code is 200 (OK)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>AJAX Read File</title>
</head>
<body>
<h2>AJAX: Read File Asynchronously</h2>
<button onclick="readFile()">Read File</button>
<div id="output" style="margin-top:15px; white-space: pre-wrap; border:1px
solid #ccc; padding:10px;"></div>
<script>
function readFile() {
  const xhr = new XMLHttpRequest();
  xhr.open('GET', 'sample.txt', true);
  xhr.onreadystatechange = function() {
    if(xhr.readyState === 4) {
      if(xhr.status === 200) {
        document.getElementById('output').innerText = xhr.responseText;
      } else {
        document.getElementById('output').innerText = "Error: Unable to load file.
Status code: " + xhr.status; }
    }  };
  xhr.send();}
</script>
</body>
</html>
```

```
┌─[prajwal@parrot]─[~/Desktop/web_lab_report/prajwal_js]
└──╼ $python -m http.server
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
127.0.0.1 - - [02/Sep/2025 23:42:31] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [02/Sep/2025 23:42:39] "GET /17.html HTTP/1.1" 304 -
127.0.0.1 - - [02/Sep/2025 23:42:42] "GET /sample.txt HTTP/1.1" 304 -
127.0.0.1 - - [02/Sep/2025 23:43:29] "GET /sample.txt HTTP/1.1" 200 -
----------------------------------------
```